

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280473

Kind Code

A1

Publication Date

September 04, 2025

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COMBINATION MICROWAVE AND HOOD SYSTEM

Abstract

The present application discloses a cooking device, which includes an inner chamber, an outer cover and an air suction assembly; the inner chamber is provided with a cooking cavity; the outer cover is arranged outside the inner chamber, and a heat dissipation air duct and a fume air duct is formed between the outer cover and the inner chamber; at least part of the heat dissipation air duct is located on a side of the inner chamber, a first air inlet of the heat dissipation air duct is provided on the outer cover and arranged corresponding to the side of the inner chamber, the heat dissipation air duct communicates with the cooking cavity and the fume air duct respectively, and a set of power devices of the cooking device are placed in the heat dissipation air duct.

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Family ID: 83006658

Appl. No.: 19/214003

Filed: May 20, 2025

Foreign Application Priority Data

CN	202110224537.6	Mar. 01, 2021
CN	202110225393.6	Mar. 01, 2021

Related U.S. Application Data

parent US continuation 17477445 20210916 parent-grant-document US 12324082 child US 19214003

Publication Classification

Int. Cl.: **H05B6/64** (20060101); **A47J36/38** (20060101); **A47J37/06** (20060101); **F24C15/20** (20060101)

U.S. Cl.:

CPC **H05B6/6423** (20130101); **A47J36/38** (20130101); **A47J37/0629** (20130101); **F24C15/2007** (20130101);

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] The present application is a continuation application of U.S. application Ser. No. 17/477,445, filed on Sep. 16, 2021, which claims benefit of Chinese Application No. 202110225393.6 and No. 202110224537.6, both filed on Mar. 1, 2021, the contents of which are hereby incorporated by reference in its entirety.

FIELD

[0002] The present application relates to the field of household appliances; in particular, the present application relates to a cooking device.

BACKGROUND

[0003] This section provides only background information related to the present disclosure, which is not necessarily the prior art.

[0004] Cooking devices such as OTR (over the range) cooking devices are usually suspended above the stove. The cooking device uses its own fume suction function to suction fume generated during the cooking process of the stove, and the cooking device uses its own microwave heating function to heat food in a cooking cavity.

[0005] The cooking device includes an air duct system, which includes a heat dissipation air duct and an fume air duct. When the cooking device is working, the air duct system has high wind noise and poor smoothness of airflow, which reduces the user experience.

SUMMARY

[0006] An object of the present application is to at least solve the problem of how to reduce wind noise and improve the smoothness of airflow.

[0007] The present application proposes a cooking device, which includes: [0008] an inner chamber, which is provided with a cooking cavity; [0009] an outer cover, the outer cover being arranged outside the inner chamber, and a heat dissipation air duct and a fume air duct being formed between the outer cover and the inner chamber; in which at least part of the heat dissipation air duct is located on a side of the inner chamber, a first air inlet of the heat dissipation air duct is provided on the outer cover and arranged corresponding to the side of the inner chamber, the heat dissipation air duct communicates with the cooking cavity and the fume air duct respectively, and a set of power devices of the cooking device are placed in the heat dissipation air duct; and [0010] an air suction assembly including a first air suction member and a second air suction member, the first air suction member being arranged in the heat dissipation air duct, and the second air suction member being arranged in the fume air duct and located at a rear of the inner chamber.

[0011] According to the cooking device of the present application, the outer cover is arranged outside the inner chamber, and a space between the outer cover and the inner chamber forms a heat dissipation air duct and an fume air duct; the heat dissipation air duct communicates with the cooking cavity and the fume air duct respectively, the set of power devices and the first air suction member are both arranged in the heat dissipation air duct, and the second air suction member is

arranged in the fume air duct. When the cooking device is in use (the cooking function and the fume suction function are used at the same time), both the first air suction member and the second air suction member are activated. Under the action of the first air suction member, air enters the heat dissipation air duct, and the air entering the heat dissipation air duct dissipates heat from the set of power devices through heat exchange. A part of the air after heat exchange enters the cooking cavity and then is discharged, and the other part of the air enters the fume air duct and then is discharged. By communicating the heat dissipation air duct with the cooking cavity and the fume air duct, obstruction to the air during the flow in the heat dissipation air duct can be reduced, so that the smoothness of air flow is improved. In addition, since the heat dissipation air duct communicates with the fume air duct, a flow rate of the air can be effectively increased under a joint action of the first air suction member and the second air suction member, so that a heat dissipation effect on the set of power devices is ensured.

[0012] In addition, the cooking device according to the present application may also have the following additional features.

[0013] In some embodiments of the present application, at least part of the set of power devices is located on a side of the inner chamber.

[0014] In some embodiments of the present application, the cooking device further includes a partition assembly arranged between the outer cover and the inner chamber, and the partition assembly divides the side of the inner chamber at least into a first electrical room and a second electrical room; the first air inlet communicates with the first electrical room, the first electrical room communicates with the cooking cavity, the second electrical room and the fume air duct respectively, and the second electrical room communicates with the fume air duct.

[0015] In some embodiments of the present application, the second electrical room is located at a rear of the first electrical room.

[0016] In some embodiments of the present application, the partition assembly includes: [0017] a first partition, which is located on the side of the inner chamber, and which is provided with a first through hole communicating with the first air inlet; and [0018] a second partition, which is located on the side of the inner chamber and connected with the first partition; in which the second partition, the first partition, the first outer cover and the side of the inner chamber together enclose the first electrical room, the second partition is provided with a second through hole and a third through hole, the first electrical room communicates with the second electrical room through the second through hole, and the first electrical room communicates with the fume air duct through the third through hole.

[0019] In some embodiments of the present application, the partition assembly further includes a third partition arranged at the rear of the inner chamber; the third partition, the inner chamber and the outer cover together enclose the second electrical room, the third partition is provided with a fourth through hole, and the second electrical room communicates with the fume air duct through the fourth through hole.

[0020] In some embodiments of the present application, the cooking device further includes a support plate with a second air inlet, and the support plate is arranged at the rear of the inner chamber; the support plate, the inner chamber, the third partition and the outer cover together enclose a first air duct portion of the fume air duct; the second air suction member is arranged in the first air duct portion and communicates with the outside through the second air inlet; and the first air duct portion communicates with the first electrical room and the second electrical room respectively.

[0021] In some embodiments of the present application, the cooking device further includes a first air guide member; the first air guide member is arranged at the rear of the inner chamber, and the first air guide member and the second air suction member are respectively located on two opposite sides of the third partition; the first air guide member, the inner chamber, the third partition and the outer cover together enclose a second air duct portion of the fume air duct, and the first air duct

portion communicates with the second air duct portion through the fourth through hole of the third partition.

[0022] In some embodiments of the present application, the cooking device further includes a second air guide member, which is arranged at a top of the inner chamber; the inner chamber, the second air guide member and the outer cover together enclose a third air duct portion of the fume air duct, the third air duct portion communicates with the second air duct portion, and the outer cover is provided with a first air outlet that communicates with the third air duct portion.

[0023] In some embodiments of the present application, the inner chamber is provided with an air intake hole and an air exhaust hole that communicate with the cooking cavity respectively, the cooking cavity communicates with the first electrical room through the air intake hole, and the cooking cavity communicates with the outside through the air exhaust hole.

[0024] In some embodiments of the present application, the cooking device further includes a third air guide member which is arranged between the inner chamber and the outer cover, and the air exhaust hole communicates with a second air outlet on the outer cover through the third air guide member.

[0025] In some embodiments of the present application, the first partition and the outer cover together enclose a third electrical room; the first electrical room, the second electrical room and the third electrical room communicate with each other in sequence, and a plurality of power devices of the set of power devices are dispersedly placed in the first electrical room, the second electrical room and the third electrical room.

[0026] In some embodiments of the present application, a third air outlet is provided on the outer cover, and the third electrical room communicates with the outside through the third air outlet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The drawings are only used for the purpose of illustrating the embodiments, and should not be considered as a limitation to the present application. Moreover, throughout the drawings, identical components are denoted by identical reference signs. In the drawings:

[0028] FIG. 1 schematically shows a structural view of a cooking device according to an embodiment of the present application;

[0029] FIG. 2 is a schematic exploded structural view of the cooking device shown in FIG. 1;

[0030] FIG. 3 is a schematic structural view of a first embodiment of the cooking device shown in FIG. 2 when in a first posture (partial structure is shown, in which black solid arrows indicate an air flow direction of a heat dissipation air duct, and black hollow arrows indicate an air flow direction of a fume air duct);

[0031] FIG. 4 is a schematic structural view of the cooking device shown in FIG. 2 in a second posture (partial structure is shown, in which black solid arrows indicate the air flow direction of the heat dissipation air duct, and black hollow arrows indicate the air flow direction of the fume air duct);

[0032] FIG. 5 is a schematic structural view of the cooking device shown in FIG. 2 in a third posture (partial structure is shown, in which black solid arrows indicate the air flow direction of the heat dissipation air duct, and black hollow arrows indicate the air flow direction of the fume air duct);

[0033] FIG. 6 is a schematic structural view of a second embodiment of the cooking device shown in FIG. 2 (partial structure is shown, and a set of power devices are distributed in a first electrical room, a second electrical room and a third electrical room, in which black solid arrows indicate the air flow direction of the heat dissipation air duct, and black hollow arrows indicate the air flow direction of the fume air duct).

LIST OF REFERENCE SIGNS

[0034] **100**: cooking device; [0035] **10**: inner chamber; **11**: cooking cavity; **12**: air intake hole; **13**: air exhaust hole; [0036] **20**: outer cover; **21**: side plate; **211**: first air inlet; **212**: first air outlet; **213**: second air outlet; **22**: rear plate; [0037] **30**: partition assembly; **31**: first partition; **311**: first through hole; **32**: second partition; **321**: second through hole; **322**: third through hole; **33**: third partition; **331**: fourth through hole; [0038] **40**: suction assembly; **41**: first air suction member; **42**: second air suction member; [0039] **50**: support plate; **51**: second air inlet; [0040] **60**: first air guide member; [0041] **70**: second air guide member; [0042] **80**: third air guide member; [0043] **90**: door assembly; [0044] **101**: heat dissipation air duct; **1011**: first electrical room; **1012**: second electrical room; **1013**: third electrical room; [0045] **102**: fume air duct; **1021**: first air duct portion; **1022**: second air duct portion; **1023**: third air duct portion; [0046] **103**: control panel; [0047] **104**: a set of power devices.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0048] Hereinafter, exemplary embodiments of the present disclosure will be described in more detail with reference to the accompanying drawings. Although the exemplary embodiments of the present disclosure are shown in the drawings, it should be understood that the present disclosure may be implemented in various forms and should not be limited by the embodiments set forth herein.

[0049] It should be understood that the terms used herein are only for the purpose of describing specific exemplary embodiments, and are not intended to be limitative. Unless clearly indicated otherwise in the context, singular forms “a”, “an”, and “said” as used herein may also mean that plural forms are included. Terms “include”, “comprise”, “contain” and “have” are inclusive and therefore indicate the existence of the stated features, steps, operations, elements and/or components, but do not exclude the existence or addition of one or more other features, steps, operations, elements, components, and/or combinations thereof. The method steps, processes, and operations described herein should not be interpreted as requiring them to be executed in the specific order described or illustrated, unless the order of execution is clearly indicated. It should also be understood that additional or alternative steps may be used.

[0050] Although terms “first”, “second”, “third” and the like may be used herein to describe multiple elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may only be used to distinguish one element, component, region, layer or section from another region, layer or section. Unless clearly indicated in the context, terms such as “first”, “second” and other numerical terms do not imply an order or sequence when they are used herein. Therefore, the first element, component, region, layer or section discussed below may be referred to as a second element, component, region, layer or section without departing from the teachings of the exemplary embodiments.

[0051] For ease of description, spatial relative terms may be used herein to describe the relationship of one element or feature relative to another element or feature as shown in the drawings. These relative terms are, for example, “inner”, “outer”, “inside”, “outside”, “below”, “under”, “above”, “over”, etc. These spatial relative terms are intended to include different orientations of the device in use or operation in addition to the orientation depicted in the drawings. For example, if the device in the figure is turned over, then elements described as “below other elements or features” or “under other elements or features” will be oriented as “above the other elements or features” or “over the other elements or features”. Thus, the exemplary term “below” may include orientations of both above and below. The device can be otherwise oriented (rotated by 90 degrees or in other directions), and the spatial relationship descriptors used herein will be explained accordingly.

[0052] As shown in FIGS. **1** to **5**, according to an embodiment of the present application, a cooking device **100** is proposed. The cooking device **100** includes an inner chamber **10**, an outer cover **20**, and a suction assembly **40**. The inner chamber **10** is provided with a cooking cavity **11**, the outer

cover **20** is arranged outside the inner chamber **10**, and a heat dissipation air duct **101** and a fume air duct **102** are formed between the outer cover **20** and the inner chamber **10**. At least part of the heat dissipation air duct **101** is located on a side of the inner chamber **10**, a first air inlet **211** of the heat dissipation air duct **101** is provided on the outer cover **20** and arranged corresponding to the side of the inner chamber **10**, the heat dissipation air duct **101** communicates with the cooking cavity **11** and the fume air duct **102** respectively, and a set of power devices **104** of the cooking device **100** are placed in the heat dissipation air duct **101**. The air suction assembly **40** includes a first air suction member **41** and a second air suction member **42**, the first air suction member **41** is arranged in the heat dissipation air duct **101**, and the second air suction member **42** is arranged in the fume air duct **102** and located at a rear of the inner chamber **10**.

[0053] Specifically, the outer cover **20** is arranged outside the inner chamber **10**, and a space between the outer cover **20** and the inner chamber **10** forms the heat dissipation air duct **101** and the fume air duct **102**. The heat dissipation air duct **101** communicates with the cooking cavity **11** and the fume air duct **102** respectively, the set of power devices **104** and the first air suction member **41** are both arranged in the heat dissipation air duct **101**, and the second air suction member **42** is arranged in the fume air duct **102**. When the cooking device **100** is in use (the cooking function and the fume suction function are used at the same time), both the first air suction member **41** and the second air suction member **42** are activated. Under the action of the first air suction member **41**, air enters the heat dissipation air duct **101**, and the air entering the heat dissipation air duct **101** dissipates heat from the set of power devices **104** through heat exchange. A part of the air after heat exchange enters the cooking cavity **11** and then is discharged, and the other part of the air enters the fume air duct and then is discharged. By communicating the heat dissipation air duct **101** with the cooking cavity **11** and the fume air duct, obstruction to the air during the flow in the heat dissipation air duct **101** can be reduced, so that the smoothness of air flow is improved. In addition, since the heat dissipation air duct communicates with the fume air duct, a flow rate of the air can be effectively increased under a joint action of the first air suction member **41** and the second air suction member **42**, so that a heat dissipation effect on the set of power devices **104** is ensured.

[0054] It should be understood that at least part of the heat dissipation air duct **101** is arranged on the side of the inner chamber **10**; at this time, at least part of the set of power devices **104** can be arranged in the air duct of the heat dissipation air duct **101** that is located on the side of the inner chamber **10**, and at the same time, the arrangement of the second air suction member **42** at the rear of the inner chamber **10** can prevent a top space of the inner chamber **10** from being occupied, and achieving the purpose of reducing the height of the cooking device **100** and meeting user's installation requirements in a small space.

[0055] It should be pointed out that in the present application, the cooking device **100** is provided with a pick-and-place port, which communicates with the cooking cavity **11** of the inner chamber **10**. After installation of the cooking device **100** is completed, when the user faces the pick-and-place port, a side of the inner chamber **10** that is close to the user is a front of the inner chamber **10**, a side of the inner chamber **10** that is away from the user is a rear of the inner chamber **10**, a side of the inner chamber **10** that is located on the user's left hand is a left side, and a side of the inner chamber **10** that is located on the user's right hand is a right side. The left and right sides are collectively referred to as the sides of the inner chamber **10**. A side of the inner chamber **10** that is away from the stove is a top of the inner chamber **10**, and a side of the inner chamber **10** that is close to the stove is a bottom of the inner chamber **10**. A distance from the left side to the right side of the cooking device **100** is a length of the cooking device **100**, a distance from the top to the bottom of the cooking device **100** is a height of the cooking device **100**, and a distance from the front to the rear of the cooking device **100** is a width of the cooking device **100**.

[0056] In addition, as shown in FIGS. **1** and **2**, the cooking device **100** further includes a door assembly **90** and a control panel **103**. The door assembly **90** is pivotally fitted with a front plate of

the inner chamber **10**, and the door assembly **90** is opened or closed so that opening or closing of the pick-and-place port is realized; the control panel **103** is also provided on the front plate of the inner chamber **10**, and the user can control the cooking device **100** by manipulating the control panel **103**.

[0057] In addition, as shown in FIGS. **1** and **2**, the outer cover **20** includes a side plate **21** and a rear plate **22**. The side plate **21** is arranged on the left side, the top, and the right side of the inner chamber **10**, and is connected and fixed to the front plate and a bottom plate of the inner chamber **10** through fasteners such as screws; the rear plate **22** is arranged at the rear of the inner chamber **10**, and is connected and fixed to the bottom plate and the side plate **21** of the inner chamber **10** through fasteners such as screws. By dividing the outer cover **20** into two parts, i.e., the side plate **21** and the rear plate **22**, when the cooking device **100** needs to be repaired, it is only required to remove the side plate **21** or the rear plate **22**, and improving the maintenance efficiency of the cooking device **100**. Moreover, by dividing the outer cover **20** into the two parts of the side plate **21** and the rear plate **22**, processing and manufacturing are facilitated, and the manufacturing cost of the cooking device **100** can be effectively reduced.

[0058] It is further understood that at least part of the set of power devices **104** is located on the side of the inner chamber **10**. By arranging at least part of the set of power devices **104** on the side of the inner chamber **10**, the set of power devices **104** is further prevented from occupying the top space of the inner chamber **10**, so that the height of the cooking device **100** is reduced, and meeting user's installation requirements in a small space.

[0059] Further, as shown in FIGS. **2** to **4**, the cooking device **100** further includes a partition assembly **30** arranged between the outer cover **20** and the inner chamber **10**, and the partition assembly **30** divides the side of the inner chamber **10** at least into a first electrical room **1011** and a second electrical room **1012**. The first air inlet **211** communicates with the first electrical room **1011**. The first electrical room **1011** communicates with the cooking cavity **11**, the second electrical room **1012** and the fume air duct **102** respectively. The second electrical room **1012** communicates with the fume air duct **102**. Specifically, the partition assembly **30** divides the side (the left side or the right side) of the inner chamber **10** into the first electrical room **1011** and the second electrical room **1012**. The first electrical room **1011** communicates with the second electrical room **1012**, the cooking cavity **11** and the fume air duct **102** respectively. When the cooking device **100** is activated, the first air suction member **41** and the second air suction member **42** are activated, so that the outside air enters the first electrical room **1011** through the first air inlet **211**, and the air entering the first electrical room **1011** exchanges heat with the power devices of the set of power devices **104** in the first electrical room **1011** to lower the temperature of the power devices; the air passing through the first electrical room **1011** enters the cooking cavity **11**, the second electrical room **1012**, and the fume air duct **102**, respectively. The air entering the cooking cavity **11** from the first electrical room **1011** is discharged through an exhaust structure of the cooking cavity **11**. The air entering the second electrical room **1012** from the first electrical room **1011** exchanges heat with the power devices located in the second electrical room **1012**, and the air after the heat exchange enters the fume air duct **102**. The air entering the fume air duct **102** from the second electrical room **1012** is discharged through the fume air duct **102**, and the air entering the fume air duct **102** from the first electrical room **1011** is also discharged through the fume air duct **102**.

[0060] By arranging multiple power devices of the set of power devices **104** in multiple electrical rooms respectively, the layout of the set of power devices **104** in the heat dissipation air duct **101** is made more reasonable, which reduces the obstruction to the air, and reducing wind noise in the air flow process and improving the smoothness in the air flow process.

[0061] Furthermore, as shown in FIGS. **2** to **4**, the second electrical room **1012** is located at the rear of the first electrical room **1011**. Specifically, the first electrical room **1011** is located on the side (the left side or the right side) of the inner chamber **10**, and the second electrical room **1012** is arranged corresponding to the first electrical room **1011** and is located at the rear of the first

electrical room **1011**. Therefore, the length of the cooking device **100** is effectively reduced, and the structure of the cooking device **100** is made more compact, and reducing the space occupied by the cooking device **100** during the installation, and meeting user's installation requirements in a small space.

[0062] Further, as shown in FIGS. 2 to 4, the partition assembly **30** includes a first partition **31** and a second partition **32**. The first partition **31** is located on the side of the inner chamber **10**, and the first partition **31** is provided with a first through hole **311** communicating with the first air inlet **211**. The second partition **32** is located on the side of the inner chamber **10** and is connected to the first partition **31**; the second partition **32**, the first partition **31**, the first outer cover **20** and the side of the inner chamber **10** together enclose the first electrical room **1011**. The second partition **32** is provided with a second through hole **321** and a third through hole **322**, the first electrical room **1011** communicates with the second electrical room **1012** through the second through hole **321**, and the first electrical room **1011** communicates with the fume air duct **102** through the third through hole **322**. Specifically, the first partition **31** is provided on the side of the inner chamber **10** and extends in a width direction of the cooking device **100**, and the second partition **32** is provided on the side of the inner chamber **10** and extends in a height direction of the cooking device **100**. The second partition **32** is connected perpendicularly to the first partition **31**, and the first electrical room **1011** formed by being enclosed by the second partition **32**, the first partition **31**, the first outer cover **20** and the side of the inner chamber **10** altogether is a part of the heat dissipation air duct **101**, so that the overall structure is simple and the manufacturing cost is low. In addition, the communication between the first electrical room **1011** and the outside is achieved through the first through hole **311** on the first partition **31**, and the communication between the first electrical room **1011** and the fume air duct **102** is achieved through the second through hole **321** and the third through hole **322** on the second partition **32**, which is simple in structure and facilitates processing and manufacturing.

[0063] It should be pointed out that as shown in FIGS. 2 to 4, the first electrical room **1011** is located on the side of the inner chamber **10**, the first partition **31** and the second partition **32** are both flat plates, and the first partition **31** and the second partition **32** are arranged perpendicular to each other; the direction of the second through hole **321** is consistent with the direction of the third through hole **322**, and the direction of the first through hole **311** is perpendicular to the direction of the second through hole **321** and the direction of the third through hole **322** respectively. When the air enters the first electrical room **1011** through the first through hole **311**, it flows in a direction toward the second partition **32** under the action of the first air suction member **41**; by providing the second through hole **321** and the third through hole **322** on the second partition **32**, when the air reaches the second partition **32**, it can directly enter the second electrical room **1012** through the second through hole **321**, and directly enter the fume air duct **102** through the third through hole **322**, which reduces the impact of the second partition **32** on the air, so that the smoothness of the air is improved, and the wind noise in the air flow process is also reduced.

[0064] In addition, the first through hole **311** is provided at a position far away from the second partition **32**, which can increase the path of air in the first electrical room **1011**, so that heat of the various power devices of the set of power devices **104** located in the first electrical room **1011** can be sufficiently dissipated, and it is further ensured that the set of power devices **104** operate stably and efficiently.

[0065] In addition, as shown in FIGS. 2 to 4, the second through hole **321** is of a first mesh structure, and the third through hole **322** is of a second mesh structure. By setting the second through hole **321** as the first mesh and setting the third through hole **322** as the second mesh, the smoothness of the air flow is improved, and the wind noise in the air flow process is reduced.

[0066] Further, as shown in FIGS. 2 to 4, the partition assembly **30** further includes a third partition **33**, the third partition **33** is arranged at the rear of the inner chamber **10**, and the third partition **33**, the inner chamber **10** and the outer cover **20** together enclose the second electrical room **1012**; the

third partition **33** is provided with a fourth through hole **331**, and the second electrical room **1012** communicates with the fume air duct **102** through the fourth through hole **331**. Specifically, the third partition **33** is arranged at the rear of the inner chamber **10** and extends in a length direction of the cooking device **100**, and one end of the third partition **33** protrudes out of the side of the inner chamber **10**; the part of the third partition **33** that protrudes out of the inner chamber **10** is connected perpendicularly to the second partition **32**, and encloses the second electrical room **1012** together with the inner chamber **10** and the outer cover **20**. The second electrical room **1012** communicates with the fume air duct **102** through the fourth through hole **331**. The formation of the second electrical room **1012** is simple in structure, which facilitates processing and manufacturing, and can effectively reduce the manufacturing cost of the cooking device **100**.

[0067] It should be pointed out that the second through hole **321** on the second partition **32** communicates with the second electrical room **1012**, the third through hole **322** on the third partition **33** communicates with the fume air duct **102**, and the direction of the second through hole **321** is perpendicular to the direction of the third through hole **322**. The air in the first electrical room **1011** enters the second electrical room **1012** through the second through hole **321**, and then enters the fume air duct **102** through the third through hole **322**. Through the arrangement of the second through hole **321** and the third through hole **322**, the smoothness of the air flow can be further improved, and the wind noise in the air flow process is effectively reduced.

[0068] In addition, the third partition **33** is a plate-shaped member, and the third partition **33**, which is a plate-shaped member, has a simple structure, which facilitates processing and manufacturing, and leads to a low manufacturing cost.

[0069] Further, as shown in FIGS. 2 to 4, the cooking device **100** further includes a support plate **50** having a second air inlet **51**, the support plate **50** is arranged at the rear of the inner chamber **10**, and the support plate **50**, the inner chamber **10**, the third partition **33** and the outer cover **20** together enclose a first air duct portion **1021** of the fume air duct **102**. The second air suction member **42** is arranged in the first air duct portion **1021** and communicates with the outside through the second air inlet **51**. The first air duct portion **1021** communicates with the first electrical room **1011** and the second electrical room **1012**, respectively. Specifically, the outer cover **20** is arranged outside the inner chamber **10**, and the third partition **33** and the support plate **50** are both arranged between the outer cover **20** and the inner chamber **10**. The third partition **33** is arranged at the rear of the inner chamber **10** and extends in the length direction of the cooking device **100**. The support plate **50** is also arranged at the rear of the inner chamber **10**, and the support plate **50** is arranged under the third partition **33** in a parallel and spaced-apart manner. The second air suction member **42** is arranged on the support plate **50** and located between the support plate **50** and the third partition **33**. The third partition **33**, the support plate **50**, the inner chamber **10** and the outer cover **20** together enclose the first air duct portion **1021** of the fume air duct **102**, and the first air duct portion **1021** communicates with the outside through the second air inlet **51** on the support plate **50**. When it is necessary to suction fume from the stove below the cooking device **100**, the second air suction member **42** is activated. Under the action of the first air suction member **41**, the fume enters the first air duct portion **1021** through the second air inlet **51** and enters a fume exhaust duct along the fume air duct **102** to realize the discharging of fume. By arranging the second air suction member **42** in the first air duct portion **1021**, the height of the cooking device **100** can be effectively reduced while ensuring normal suctioning of fume, and reducing the space required for installing the cooking device **100** and meeting user's installation requirements in a small space.

[0070] It should be pointed out that in the present application, the support plate **50** is flush with the bottom plate of the inner chamber **10**, and the third partition **33** is arranged above the support plate **50** in a spaced-apart manner. The third partition **33**, the support plate **50**, the inner chamber **10** and the outer cover **20** together enclose the first air duct portion **1021** of the fume air duct **102**, and the first air duct portion **1021** is located below the rear of the inner chamber **10**, which further prevents

the second air suction member **42** from occupying the top space of the inner chamber **10**, so that the height of the cooking device **100** can be reduced to meet user's installation requirements in a small space.

[0071] In addition, as shown in FIGS. **3** and **4**, in the present application, the second air suction member **42** is a vortex fan, the second air inlet **51** on the support plate **50** is a first grille structure, and the second air inlet **51** is arranged close to an inlet of the vortex fan to ensure the suctioning effect of the vortex fan on the fume, so that the discharging effect of the fume is guaranteed.

[0072] In addition, as shown in FIG. **4**, the number of the second air suction members **42** may be plural, and the plurality of second air suction members **42** are arranged at intervals in the first air duct portion **1021**. By providing the plurality of second air suction members **42**, the ability of the cooking device **100** to suction fume is improved, and the effect of the cooking device **100** on discharging the fume is further improved.

[0073] Further, as shown in FIGS. **2** to **4**, the cooking device **100** further includes a first air guide member **60**, the first air guide member **60** is arranged at the rear of the inner chamber **10**, and the first air guide member **60** and the second air suction member **42** are respectively located on opposite sides of the third partition **33**. The first air guide member **60**, the inner chamber **10**, the third partition **33** and the outer cover **20** together enclose a second air duct portion **1022** of the fume air duct **102**. The first air duct portion **1021** communicates with the second air duct portion **1022** through the fourth through hole **331** of the third partition **33**. Specifically, the first air guide member **60** is arranged at the rear of the inner chamber **10**, and the first air guide member **60** and the second air suction member **42** are located on both sides of the third partition **33**. The first air guide member **60**, the third partition **33**, the inner chamber **10** and the outer cover **20** together enclose the second air duct portion **1022** of the fume air duct **102**. The first air duct portion **1021** communicates with the second air duct portion **1022** through the fourth through hole **331** of the third partition **33**. When it is necessary to suction fume from the stove below the cooking device **100**, the second air suction member **42** is activated. Under the action of the second air suction member **42**, the fume enters the first air duct portion **1021** through the second air inlet **51**, then enters the second air duct portion **1022** through the fourth through hole **331**, and finally enter the fume exhaust duct along the fume air duct **102** to realize the discharging of fume. By providing the second air duct portion **1022**, a guidance to the fume is realized, so that the fume is discharged according to a preset path, and ensuring the discharging effect of the fume.

[0074] It should be pointed out that as shown in FIG. **4**, the first air guide member **60** is a first air guide plate, and the first air guide plate has a simple structure, and is easy to process and manufacture, which can effectively ensure the manufacturing cost of the first air guide member **60**. In addition, the first air guide plate is arranged obliquely and at an angle to the flow direction of the air flow, and ensuring a better guidance to the air, and further improving the smoothness of the air flow.

[0075] In addition, as shown in FIG. **4**, when the number of the second air suction members **42** is plural, the number of the fourth through holes **331** on the third partition **33** and the number of the first air guide members **60** each have a one-to-one correspondence to the number of the second air suction members. That is, each second air suction member **42** corresponds to one fourth through hole **331** and one first air guide member **60**, to ensure the effective guidance to the fume, further improving the effect of discharging the fume.

[0076] Further, as shown in FIGS. **2** to **4**, the cooking device **100** further includes a second air guide member **70**, the second air guide member **70** is arranged at the top of the inner chamber **10**, and the inner chamber **10**, the second air guide member **70** and the outer cover **20** together enclose a third air duct portion **1023** of the fume air duct **102**; the third air duct portion **1023** communicates with the second air duct portion **1022**, and the outer cover **20** is provided with a first air outlet **212** communicating with the third air duct portion **1023**. Specifically, the third air duct portion **1023** is located at the top of the inner chamber **10**, and the third air duct portion **1023**, the second air duct

portion **1022** and the first air duct portion **1021** communicate with each other in sequence to form the fume air duct **102**; the fume air duct **102** covers the rear and the top of the inner chamber **10**, and the third air duct portion **1023** formed by the second air guide member **70**, the inner chamber **10** and the outer cover **20** is arranged at the top of the inner chamber **10**; by controlling the height of the second air guide member **70**, the height of the cooking device **100** can be effectively reduced while ensuring the discharging of the fume, so that user's installation requirements in a small space can be met.

[0077] It should be pointed out that the second air guide member **70** is a second air guide plate, and the number of the second air guide plates is two; the two second air guide plates are arranged at the top of the inner chamber **10** in a parallel and spaced-apart manner, a space between the two second air guide plates constitutes the third air duct portion **1023**, and the first air outlet on the outer cover **20** communicates with the third air duct portion **1023** and is located away from the second air duct portion **1022**.

[0078] In other embodiments, an outlet of the second air duct portion **1022** directly communicates with the fume exhaust duct, and there is no need to provide the second air guide member **70**, so that the manufacturing cost of the cooking device **100** is further reduced, and at the same time, the height of the cooking device **100** can be further reduced.

[0079] Further, as shown in FIGS. **4** and **5**, the inner chamber **10** is provided with an air intake hole **12** and an air exhaust hole **13** communicating with the cooking cavity **11** respectively, the cooking cavity **11** communicates with the first electrical room **1011** through the air intake hole **12**, and the cooking cavity **11** communicates with the outside through the air exhaust hole **13**. Specifically, the cooking cavity **11** communicates with the first electrical room **1011** through the air intake hole **12**, and communicates with the outside through the air exhaust hole **13**. The air in the first electrical room **1011** enters the cooking cavity **11** through the air intake hole **12** and then takes water vapor or fume or the like in the cooking cavity **11** away from the cooking cavity **11** through the air exhaust hole **13**, and ensuring a good cooking environment in the cooking cavity **11** and improving the cooking quality of food.

[0080] It should be pointed out that the air intake hole **12** is provided on a side wall of the cooking cavity **11** that corresponds to the first electrical room **1011**, and the air exhaust hole **13** is provided on the top wall of the cooking cavity **11** and is located away from the air intake hole **12**. The arrangement of the air intake hole **12** and the air exhaust hole **13** can effectively ensure the effect of discharging water vapor or fume or the like in the cooking cavity **11**.

[0081] In some embodiments of the present application, as shown in FIGS. **2** to **4**, the cooking device **100** further includes a third air guide member **80**. The third air guide member **80** is arranged between the inner chamber **10** and the outer cover **20**, and the air exhaust hole **13** communicates with a second air outlet **213** on the outer cover **20** through the third air guide member **80**. Specifically, the third air guide member **80** communicates the air exhaust hole **13** with the second air outlet **213** on the outer cover **20** to prevent water vapor or fume or the like from reaching between the inner chamber **10** and the outer cover **20**, and ensuring the effect of discharging water vapor or fume or the like in the cooking cavity **11** and improving the sanitary conditions in the cooking cavity **11**.

[0082] Further, as shown in FIGS. **3** and **4**, the first partition **31** and the outer cover **20** together enclose a third electrical room **1013**. The first electrical room, **1011**, the second electrical room **1012** and the third electrical room **1013** communicate with each other in sequence, and a plurality of power devices of the set of power devices **104** are dispersedly placed in the first electrical room **1011**, the second electrical room **1012** and the third electrical room **1013**.

[0083] Specifically, the partition assembly **30** includes the first partition **33**, the second partition **31**, and the third partition **32**. The third partition **33** is arranged at the rear of the inner chamber **10** and extends in the length direction of the cooking device **100**. One end of the third partition **33** protrudes out of the side of the inner chamber **10** (which may either be the left side or the right

side), the first partition **31** and the second partition **32** are both arranged on the side of the inner chamber **10**, and the part of the third partition **33** that protrudes out of the side of the inner chamber **10** defines the first electrical room **1011**, the second electrical room **1012** and the third electrical room **1013** by division. The plurality of power devices of the set of power devices **104** may be dispersedly placed in these three electrical rooms, and reducing the space occupied by the installation of the set of power devices **104**, thus making the structure of the cooking device **100** more compact, reducing the volume of the cooking device **100**, and reducing the space required for installing the cooking device **100**.

[0084] It should be pointed out that the first partition **31** extends in the width direction of the cooking device **100** and is arranged close to the top of the inner chamber **10**, the second partition **32** extends in the height direction of the cooking device **100**, in which one end of the second partition **32** abuts against the first partition **31** and the other end abuts against the bottom plate of the inner chamber **10**, and one end of the third partition **33** that protrudes out of the side of the inner chamber **10** cooperates with the second partition **32** to form the first electrical room **1011**, the second electrical room **1012** and the third electrical room **1013**, in which the first electrical room **1011** and the third electrical room **1013** are arranged up and down, and the third electrical room **1013** is located at the rear of the first electrical room **1011** and the second electrical room **1012**. The arrangement of the first electrical room **1011**, the second electrical room **1012** and the third electrical room **1013** can further improve the compactness of the cooking device **100**, so that the volume of the cooking device **100** is further reduced, and the space required for installing the cooking device **100** is further reduced.

[0085] In addition, as shown in FIG. 4 or FIG. 6, the set of power devices **104** typically includes: a magnetron, a transformer, a filter, and a capacitor. Power devices (such as the magnetron, the transformer, etc.) with a larger volume are arranged in the first electrical room **1011**, power devices (such as the capacitor, etc.) with a medium volume may be placed in the second electrical room **1012**, and power devices (such as the filter, etc.) with a small volume may be placed in the third electrical room **1013**, and making the overall layout more reasonable, to achieve the purpose of further reducing the cooking device **100**.

[0086] In addition, in other embodiments of the present application, as shown in FIG. 6, the third partition **33** is not provided with the fourth through hole **331**, the second electrical room **1012** communicates with the first electrical room **1011** and the third electrical room **1013** respectively, and the first electrical room **1011**, the second electrical room **1012** and the third electrical room **1013** constitute at least part of the heat dissipation air duct **101**; the plurality of power devices of the set of power devices **104** are dispersedly placed in the first electrical room **1011**, the second electrical room **1012** and the third electrical room **1013**, and heat dissipation of the set of power devices **104** is realized through the operation of the first air suction member **41**.

[0087] Further, a third air outlet is provided on the outer cover **20**, and the third electrical room **1013** communicates with the outside through the third air outlet. Specifically, the third air outlet is arranged corresponding to the third electrical room **1013**. When the second air suction member **42** in the cooking device **100** is not activated, under the action of the first air suction member **41**, the outside air enter the first electrical room **1011** through the first air inlet **211** and the first through hole **311**, then enters the second electrical room **1012** from the first electrical room **1011**, and then enters the third electrical room **1013** from the second electrical room **1012**; the air in the third electrical room **1013** is discharged through the third air outlet, and realizing effective heat dissipation of the set of power devices **104**.

[0088] It should be pointed out that in the present application, the first air suction member **41** is a fan, and the fan has a simple structure and low manufacturing cost, which can effectively reduce the manufacturing cost of the cooking device.

[0089] In the present application, the above cooking device **100** is an OTR microwave oven. For

the structure of other parts of the OTR microwave oven, reference may be made to the prior art, so a detailed description thereof will be omitted in the present application.

Claims

1. A cooking device, comprising: an inner chamber, the inner chamber is provided with a cooking cavity; an outer cover, the outer cover being arranged outside the inner chamber, and a heat dissipation air duct and a fume air duct being formed between the outer cover and the inner chamber; an air suction assembly comprising a first air suction member, the first air suction member being arranged in the fume air duct and located at a rear of the inner chamber; and a set of power devices arranged in the heat dissipation air duct, and at least part of the set of power devices is located on a side of the inner chamber.
2. The cooking device according to claim 1 further comprises: a partition assembly arranged between the outer cover and the inner chamber, the partition assembly including a first partition disposed at rear of the inner chamber; and a support plate, wherein the support plate is arranged at the rear of the inner chamber, and arranged at lower part of the first partition in a spaced-apart manner; the first partition, the support plate, the inner chamber and the outer cover enclose a first air duct portion of the fume air duct, and the first air suction member is arranged in the first air duct portion and communicates with outside through a first air inlet of the support plate.
3. The cooking device according to claim 2 further comprises a first air guide, wherein the first air guide is arranged at the rear of the inner chamber; the first air guide, the first partition, the inner chamber and the outer cover enclose a second air duct portion of the fume air duct, and the second air duct portion communicates with the first air duct portion through a first through hole on the first partition.
4. The cooking device according to claim 3 further comprises a second air guide, wherein the second air guide is arranged on top of the inner chamber; the inner chamber, the second air guide and the outer cover together enclose a third air duct portion of the fume air duct; the third air duct portion communicates with the second air duct portion; and the outer cover is provided with a first air outlet communicating with the third air duct portion.
5. The cooking device according to claim 3, wherein the partition assembly further comprises: a second partition located on the side of the inner chamber; and a third partition; wherein the third partition is located at the side of the inner chamber and connected to the second partition; the third partition, the second partition, the outer cover and the side of the inner chamber together enclose a first electrical room of the heat dissipation air duct; the third partition, the first partition, the first air guide, the inner chamber and the outer cover together enclose a second electrical room of the heat dissipation air duct; the second partition and the outer cover together enclose a third electrical room of the heat dissipation air duct; the first electrical room, the second electrical room, and the third electrical room are sequentially communicated; and a plurality of power devices of the set of power devices are dispersedly arranged within the first electrical room, the second electrical room, and the third electrical room.
6. The cooking device according to claim 5, wherein a second through hole is provided on the third partition, and the first electrical room communicates with the second electrical room through the second through hole.
7. The cooking device according to claim 5, wherein a fourth through hole is provided on the second partition, and the first electrical room communicates with a second air inlet of the outer cover through the fourth through hole.
8. The cooking device according to claim 5, wherein the air suction assembly further comprises a second air suction member, the second air suction member is arranged in the first electrical room.
9. The cooking device according to claim 5, wherein the inner chamber is provided with an air intake hole and an air exhaust hole that communicate with the cooking cavity respectively, the

cooking cavity communicates with the first electrical room through the air intake hole, and the cooking cavity communicates with outside through the air exhaust hole.

10. The cooking device according to claim 9, wherein the cooking device further comprises a third air guide arranged between the inner chamber and the outer cover, and the air exhaust hole communicates with a third air outlet on the outer cover through the third air guide.

11. The cooking device according to claim 5, wherein a third through hole is provided on the third partition, and the second electrical room communicates with the first air duct portion through the third through hole.

12. The cooking device according to claim 5, wherein the outer cover is provided with a second air outlet, and the third electrical room communicates with outside through the second air outlet.
